

L10: Bundle Adjustment

- Minimises reprojection error in the image
- Gold standard approach
- Moving cameras for 3D reconstruction
- (So far: only image pairs)
- 3D reconstruction using $N > 2$ images
- **Joint** estimation problem

Least Squares

- Least squares approach for estimating camera poses and 3D points jointly

Key Idea

- Start with an initial guess
- Project the **estimated 3D points** into the **estimated camera images**
- Compare locations of the **projected image points** with **measured ones**
- Adjust to minimize error in images

Reprojection Error

$${}^a\mathbf{x}'_{ij} + {}^a\hat{\mathbf{v}}_{x'_{ij}} = \hat{\lambda}_{ij} {}^a\hat{\mathbf{P}}_j(\mathbf{x}_{ij}, \mathbf{p}, \mathbf{q}) \hat{\mathbf{X}}_i$$

$a = \text{"arbitrary frame"}$
 $ij = \text{point } i \text{ observed in image } j$
 $\hat{\mathbf{v}}_{x'_{ij}} = \text{corrections}$
 $\hat{\lambda}_{ij} = \text{scale factor}$
 ${}^a\hat{\mathbf{P}}_j(\mathbf{x}_{ij}, \mathbf{p}, \mathbf{q}) = \text{projection matrix (w/ non-lin. calib.)}$
 $\hat{\mathbf{X}}_i = \text{3D point}$
 $\mathbf{x}_{ij} = \text{img pix.}$
 $\mathbf{p}, \mathbf{q} = \text{project. params distortion}$

with $\sum_{x_{ij} x_{ij}}$ $i = 1, \dots, I_j; j = 1, \dots, J$
 $\sum_{x_{ij} x_{ij}} = \text{uncertainty in the image coordinates}$
 $I_j = \text{\#points in image } j$
 $J = \text{\#images}$

- Encodes camera **projection**
- Error is distance between each projected 3D and measured 2D point

- Encodes **collinearity constraint**
- **Known** data association
- **Unknown parameters:**
 - 3D locations of new points $\hat{\mathbf{X}}_i$
 - 1D scale factor $\hat{\lambda}_{ij}$
 - 6D exterior orientation
 - 5D projection parameters (interior o.)
 - Non-linear distortion parameters $\mathbf{q}$

Into Euclidian Space

- Avoid scale parameter $\hat{\lambda}_{ij}$
 - wise to move to Euclidian space

$${}^a\mathbf{x}'_{ij} + {}^a\hat{\mathbf{v}}_{x'_{ij}} = \hat{\lambda}_{ij} {}^a\hat{\mathbf{P}}_j(\mathbf{x}_{ij}, \mathbf{p}, \mathbf{q}) \hat{\mathbf{X}}_i$$

$${}^a\mathbf{x}'_{ij} + {}^a\hat{\mathbf{v}}_{x'_{ij}} = \hat{\lambda}_{ij} \hat{\mathbf{K}}(\mathbf{x}_{ij}, \hat{\mathbf{p}}, \hat{\mathbf{q}}) \hat{\mathbf{R}}_j [I_3 - \hat{\mathbf{X}}_{0j}] \hat{\mathbf{X}}_i$$

calibration
(interior
orientation)
exterior
orientation
3D point
coordinates

■ To
$${}^a\mathbf{x}'_{ij} + {}^a\hat{\mathbf{v}}_{x'_{ij}} = \frac{{}^a\hat{\mathbf{P}}_{1:2j}(\mathbf{x}_{ij}, \mathbf{p}, \mathbf{q}) \hat{\mathbf{X}}_i}{{}^a\hat{\mathbf{P}}_{3j}(\mathbf{x}_{ij}, \mathbf{p}, \mathbf{q}) \hat{\mathbf{X}}_i}$$

- Or written as an error term:

$$e(\mathbf{p}, \mathbf{q}, \hat{\mathbf{X}}_i) = {}^a\mathbf{x}'_{ij} - \frac{{}^a\hat{\mathbf{P}}_{1:2j}(\mathbf{x}_{ij}, \mathbf{p}, \mathbf{q}) \hat{\mathbf{X}}_i}{{}^a\hat{\mathbf{P}}_{3j}(\mathbf{x}_{ij}, \mathbf{p}, \mathbf{q}) \hat{\mathbf{X}}_i}$$

- For NL-LSA
- Normal equations
 - $A^T \Sigma^{-1} A \Delta \mathbf{x} = A^T \Sigma^{-1} \Delta \mathbf{l}$
 - $\Delta \mathbf{x}$: unknowns
 - $\Delta \mathbf{l}$: observations
- Yields estimate
 - $\hat{\Delta \mathbf{x}} = (A^T \Sigma^{-1} A)^{-1} A^T \Sigma^{-1} \Delta \mathbf{l}$

Optimality

- BA is **statistically optimal**
- Exploits all observations and considers uncertainties and correlations
- Exploits all available information
- Computes orientations, calibrated parameters, and point location with highest precision
- Assumes Gaussian noise
- Requires a good initial estimate

Initial Guess

- Orientation of the image pair with a direct method
- **No direct solution is known for N views**
- Computing an initial Guess
 - Compute relative orientation from 2 images
 - Compute 3D point coordinates via triangulation
 - Localize next image given so far mapped points
- **Critical issues**
 - Requires enough new points in each image overlap
 - No singular/critical configuration
 - Outlier handling is essential (later in course)

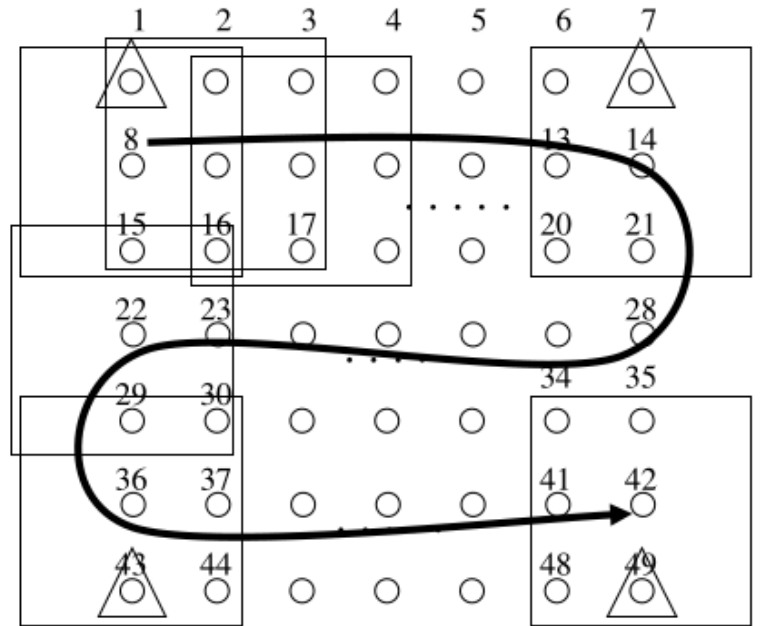
Numerics of the Bundle Adjustment

- Solving of Linear System not straightforward
 - Linear system becomes huge
 - images -> location parameters -> orientation parameters

Structure

Setup

- 3 stripes
- 7 images per stripe
- 60%/20% overlap
- 21 images
- 49 points
- 4 full CPs
- 45 new points



- Configuration
 - 6 images with 6 points
 - 15 images with 9 points
 - 171 image points yield 342 **observations**
 - 261 unknowns
 - 45 new points yield 135 **unknowns**
 - $21 \times 6 = 126$ unknowns
 - orientation parameters

$$\Delta l + v = A \Delta x = \underbrace{[C \quad B]}_A \underbrace{\begin{bmatrix} \Delta k \\ \Delta t \end{bmatrix}}_{\Delta x} = C \Delta k + B \Delta t$$

- with Δk : 3D point coordinates
- with Δt : 6D orientation parameters (cams)
- Thus, for every error equation

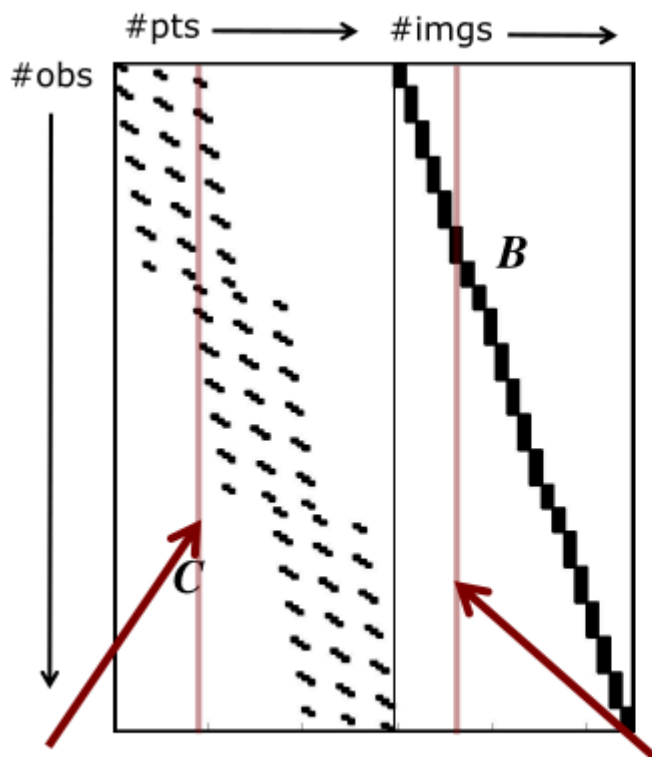
$$\Delta l_{ij} + v_{ij} = \underbrace{A_{ij}^T}_{s \times U} \Delta x = \underbrace{C_{ij}^T}_{2 \times 3} \Delta k_i + \underbrace{B_{ij}^T}_{2 \times 6} \Delta t_j$$

- Coefficient Matrix

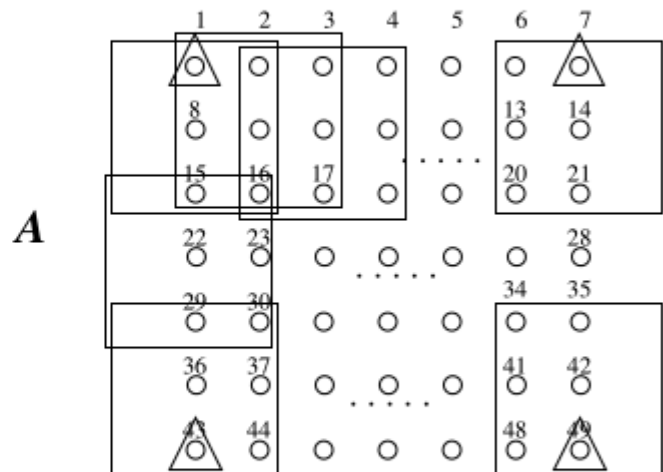
$$\circ A = \begin{bmatrix} A_{2,1}^T \\ \vdots \\ A_{ij}^T \\ \vdots \\ A_{48,21}^T \end{bmatrix}$$

$$\circ \text{ with } A_{ij}^T = [0, \dots, 0, \underbrace{C_{ij}^T}_{2 \times 3}, 0, \dots, 0 | 0, \dots, 0, \underbrace{B_{ij}^T}_{2 \times 6}, 0, \dots, 0]$$

- -> **Sparse A matrix**



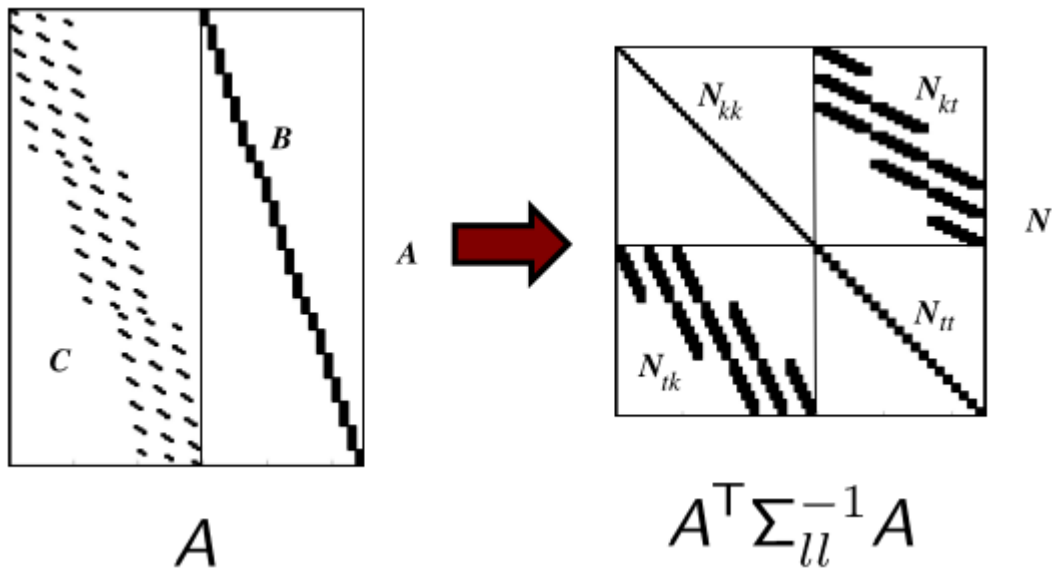
For 3 stripes with
7 images per stripe



**number of times a
point is observed**

**number of observations
(points) per image**

- Properties of C:
 - 2×3 sub-matrices C_{ij}^T
 - Each connects an image point x'_{ij} with X_i
 - one per row
 - number of non-zero 2×3 matrices per column:
 - $\#X_i$ observed
 - Properties of B:
 - 2×6 sub-matrices B_{ij}^T
 - Each connects an image point x'_{ij} with j^{th} camera orientation
 - one per row
 - number of non-zero 2×6 matrices per column:
 - $\#X_i$ observed
 - Computing Sub-Matrices: demanding
 - Usage of math tools
 - Compute Jacobians
1. Analytically
 2. Numerically



Normal Matrix

- Block diagonal covariance matrix for observations

$$\Sigma_{ll} = \text{Diag}(\Sigma_{ij})_{2 \times 2}$$

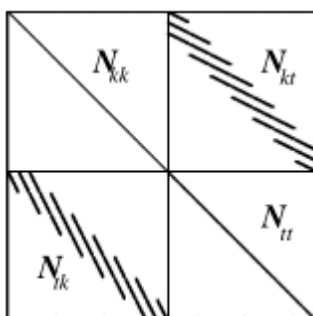
$$\begin{aligned} N &= A^T \Sigma_{ll}^{-1} A = \begin{bmatrix} C^T \\ B^T \end{bmatrix} \Sigma_{ll}^{-1} \begin{bmatrix} C & B \end{bmatrix} \\ &= \begin{bmatrix} C^T \Sigma_{ll}^{-1} C & C^T \Sigma_{ll}^{-1} B \\ B^T \Sigma_{ll}^{-1} C & B^T \Sigma_{ll}^{-1} B \end{bmatrix} = \begin{bmatrix} N_{kk} & N_{kt} \\ N_{tk} & N_{tt} \end{bmatrix} \end{aligned}$$

block-diagonal
with 3x3 blocks

sparse matrix that
reveals the connections
of images and points;
consist of 3x6 blocks

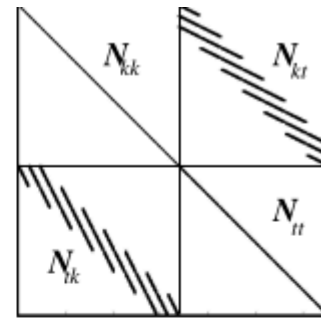
$$N = \begin{bmatrix} N_{kk} & N_{kt} \\ N_{tk} & N_{tt} \end{bmatrix}$$

Example:



block-diagonal
with 6x6 blocks

$$N = \begin{bmatrix} N_{kk} & N_{kt} \\ N_{tk} & N_{tt} \end{bmatrix}$$



$$N_{kk} = \text{Diag}(N_{k_i k_i})$$

$3N_P \times 3N_P$

$$N_{k_i k_i} = \sum_{j \in \mathcal{B}_i} C_{ij} \Sigma_{ij}^{-1} C_{ij}^T$$

all images in which
point i is observed

$$N_{tt} = \text{Diag}(N_{t_j t_j})$$

$6N_B \times 6N_B$

$$N_{t_j t_j} = \sum_{i \in \mathcal{P}_j} B_{ij} \Sigma_{ij}^{-1} B_{ij}^T$$

all points that are
observed in image j

$$N_{k_i t_j} = C_{ij} \Sigma_{ij}^{-1} B_{ij}^T$$

Orientation Parameters Only

$$N \Delta x = h$$

$$\begin{bmatrix} N_{kk} & N_{kt} \\ N_{tk} & N_{tt} \end{bmatrix} \begin{bmatrix} \Delta k \\ \Delta t \end{bmatrix} = \begin{bmatrix} h_k \\ h_t \end{bmatrix}$$

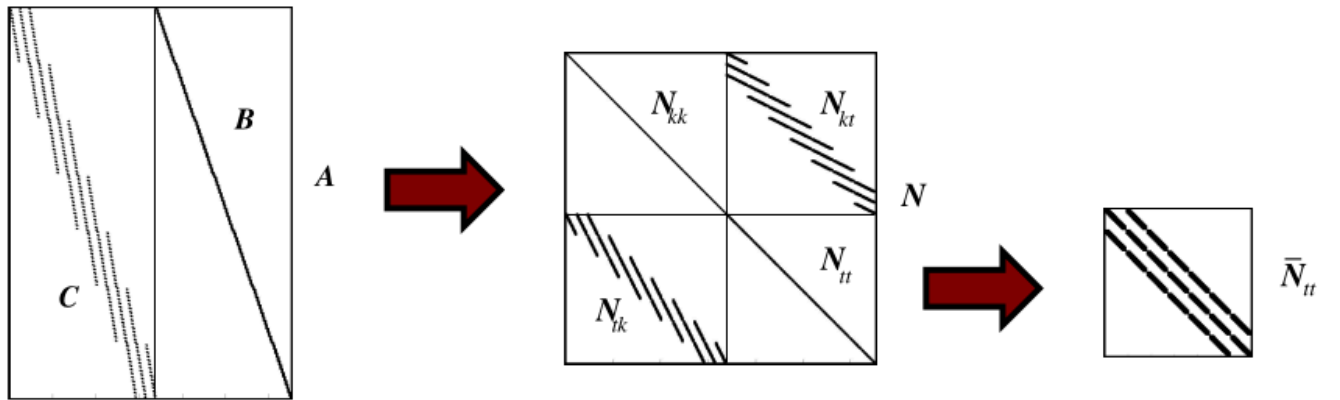
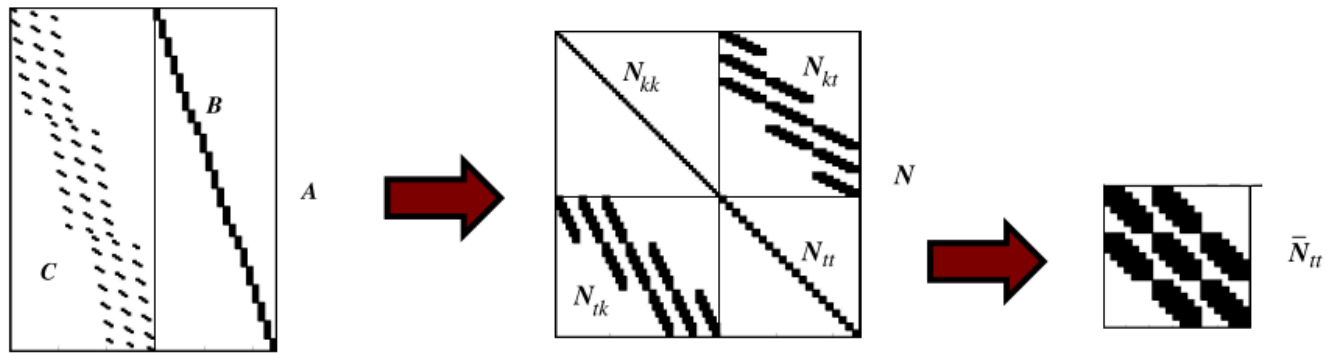
$$\begin{bmatrix} N_{kk}^{-1} & 0 \\ -N_{tk} N_{kk}^{-1} & I \end{bmatrix} \begin{bmatrix} N_{kk} & N_{kt} \\ N_{tk} & N_{tt} \end{bmatrix} \begin{bmatrix} \Delta k \\ \Delta t \end{bmatrix} = \begin{bmatrix} N_{kk}^{-1} & 0 \\ -N_{tk} N_{kk}^{-1} & I \end{bmatrix} \begin{bmatrix} h_k \\ h_t \end{bmatrix}$$

$$\begin{bmatrix} I & N_{kk}^{-1} N_{kt} \\ 0 & N_{tt} - N_{tk} N_{kk}^{-1} N_{kt} \end{bmatrix} \begin{bmatrix} \Delta k \\ \Delta t \end{bmatrix} = \begin{bmatrix} N_{kk}^{-1} h_k \\ h_t - N_{tk} N_{kk}^{-1} h_k \end{bmatrix}$$

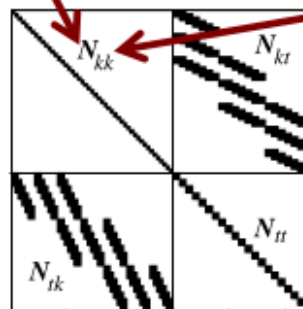
$$\bar{N}_{tt} \Delta t = \bar{h}_t$$

Reduced Normal System

- Is independent of the number of points
- Reduced system is still sparse
- $\bar{N}_{tt} \Delta t = \bar{h}_t$



• with $\bar{N}_{tt} = N_{tt} - N_{tk} N_{kk}^{-1} N_{kt}$ $\bar{h}_t = h_t - N_{tk} N_{kk}^{-1} h_k$



easy to invert!

- Solve using sparse solver

$\Delta t \rightarrow$ 3D Points

Compute point coordinates:

$$\Delta k = N_{kk}^{-1}(h_k - N_{kt}\Delta t)$$

Jacobian

- No need to construct explicitly

$$N_{k_i k_i} = \sum_{j \in \mathcal{B}_i} C_{ij} \Sigma_{ij}^{-1} C_{ij}^T$$

$$N_{k_i t_j} = C_{ij} \Sigma_{ij}^{-1} B_{ij}^T$$

$$N_{t_j t_j} = \sum_{i \in \mathcal{P}_j} B_{ij} \Sigma_{ij}^{-1} B_{ij}^T$$

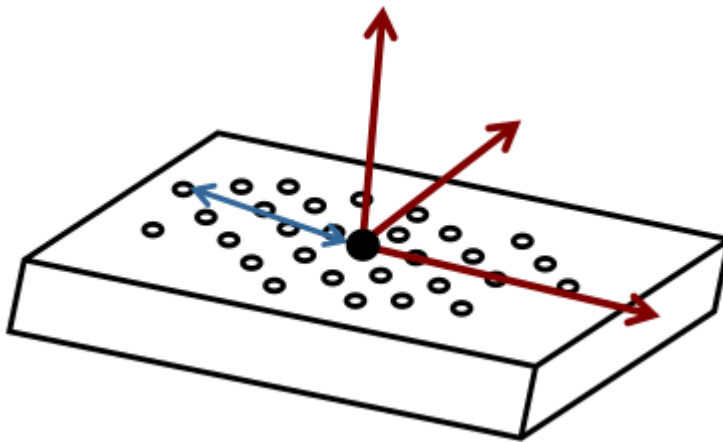
- Construct reduced system:

$$\circ \bar{N}_{tt} = N_{tt} - N_{tk} N_{kk}^{-1} N_{kt} \quad \hat{h}_t = h_t - N_{tk} N_{kk}^{-1} h_k$$

- Solve for cameras Δt with a sparse solver
- Compute for the points coordinates given Δt
 - $\Delta k = N_{kk}^{-1}(h_k - N_{kt} \Delta t)$

BA without Control Points

- No Control Points: Reference Frame is not defined
- BA only corrects geometry up to a similarity transform
 - Normal equations with rank deficiency of 7
- Gauge-freedom
- Datum has to be specified
 - Datum through additional **constraints**:
 - Center of mass of 3D point should not change (translation)
 - No change in the main directions (rotation)
 - No change in the average distance to the center of mass (scale)
 - constraint matrix $H\hat{x} = 0$
 - Jacobian H added and considered in error minimisation



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